

WEEK-2

1.To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

```
root@LAPTOP-RJFD8EBT:~/OSSP# sudo apt update && sudo apt install -y build-essential gcc git
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1227 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1687 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5772 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 4803 kB in 13s (311 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
80 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
gcc is already the newest version (4:13.2.0-7ubuntu1).
gcc set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
The following package was automatically installed and is no longer required:
git set to manually installed.
The following package was automatically installed and is no longer required:
libglapi-mesa
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 80 not upgraded.
root@LAPTOP-RJFD8EBT:~/OSSP# git init
Reinitialized existing Git repository in /root/OSSP/.git/
root@LAPTOP-RJFD8EBT:~/OSSP# mkdir -p src include obj bin docs tests scripts assets
root@LAPTOP-RJFD8EBT:~/OSSP# touch .gitignore README.md LICENSE Makefile
root@LAPTOP-RJFD8EBT:~/OSSP# touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c
root@LAPTOP-RJFD8EBT:~/OSSP# touch include/shell.h include/parser.h include/executor.h include/builtin.h
root@LAPTOP-RJFD8EBT:~/OSSP# touch docs/design.md docs/report.pdf
root@LAPTOP-RJFD8EBT:~/OSSP# touch tests/test_parser.c tests/test_executor.c
root@LAPTOP-RJFD8EBT:~/OSSP# touch scripts/run.sh assets/architecture.png
root@LAPTOP-RJFD8EBT:~/OSSP# apt install -y tree && tree -a
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
libglapi-mesa
Use 'apt autoremove' to remove it.
The following NEW packages will be installed:
tree
0 upgraded, 1 newly installed, 0 to remove and 80 not upgraded.
Need to get 47.4 kB of archives.
After this operation, 111 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 tree amd64 2.1.1-2ubuntu3.24.04.2 [47.4 kB]
Fetched 47.4 kB in 10s (4612 B/s)
Selecting previously unselected package tree.
(Reading database ... 46646 files and directories currently installed.)
Preparing to unpack .../tree_2.1.1-2ubuntu3.24.04.2_amd64.deb ...
Unpacking tree (2.1.1-2ubuntu3.24.04.2) ...
Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```

Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ..

.
├── .git
│   ├── HEAD
│   ├── branches
│   ├── config
│   ├── description
│   ├── hooks
│   │   ├── applypatch-msg.sample
│   │   ├── commit-msg.sample
│   │   ├── fsmonitor-watchman.sample
│   │   ├── post-update.sample
│   │   ├── pre-applypatch.sample
│   │   ├── pre-commit.sample
│   │   ├── pre-merge-commit.sample
│   │   ├── pre-push.sample
│   │   ├── pre-rebase.sample
│   │   ├── pre-receive.sample
│   │   ├── prepare-commit-msg.sample
│   │   ├── push-to-checkout.sample
│   │   ├── sendemail-validate.sample
│   │   └── update.sample
│   ├── info
│   │   └── exclude
│   ├── objects
│   │   ├── info
│   │   └── pack
│   └── refs
│       ├── heads
│       └── tags
├── .gitignore
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

19 directories, 37 files
root@LAPTOP-RJFD8EBT:~/OSSP# gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (c) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

root@LAPTOP-RJFD8EBT:~/OSSP# git --version
git version 2.43.0
root@LAPTOP-RJFD8EBT:~/OSSP# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        LICENSE
        Makefile
        README.md
        assets/
        docs/
        include/
        scripts/
        src/
        tests/

nothing added to commit but untracked files present (use "git add" to track)

```

1. To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the `fork()` and `exec()` system calls

```

root@LAPTOP-RJFD8EBT:~/OSSP# man 2 fork
root@LAPTOP-RJFD8EBT:~/OSSP# man 3 exec
root@LAPTOP-RJFD8EBT:~/OSSP# man 2 execve
root@LAPTOP-RJFD8EBT:~/OSSP# pstree
systemd──agetty
        ├──cron
        ├──dbus-daemon
        └──init-systemd(Ub)──SessionLeader──Relay(322)──bash──pstree
                                ├──init──{init}
                                ├──login──bash
                                └──{init-systemd(Ub)}
polkitd──3*[{polkitd}]
rsyslogd──3*[{rsyslogd}]
systemd──(sd-pam)
systemd-journal
systemd-logind
systemd-resolve
systemd-timesyn──{systemd-timesyn}
systemd-udev
unattended-upgr──{unattended-upgr}
wsl-pro-service──7*[{wsl-pro-service}]
root@LAPTOP-RJFD8EBT:~/OSSP#

```

Task-2:

Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

```
root@LAPTOP-RJFD8EBT:~/OSSP/src# nano main.c
root@LAPTOP-RJFD8EBT:~/OSSP/src# cat main.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char command[100];

    while (1) {
        printf("my_shell> ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL) {
            break;
        }

        command[strlen(command)] = '\0';

        if (strcmp(command, "exit") == 0) {
            break;
        }

        if (strlen(command) == 0) {
            continue;
        }

        pid_t pid = fork();

        if (pid < 0) {
            perror("Fork failed");
            continue;
        }

        else if (pid == 0) {
            printf("This is the Child Process.\n");
            printf("Child PID = %d\n", getpid());

            execlp(command, command, (char *)NULL);

            perror("Exec failed");
            exit(EXIT_FAILURE);
        }

        else {
            printf("This is the Parent Process.\n");
            printf("Parent PID = %d\n", getpid());
            printf("Child PID = %d\n", pid);

            wait(NULL);
        }
    }
}

exit(EXIT_FAILURE);
}

else {
    printf("This is the Parent Process.\n");
    printf("Parent PID = %d\n", getpid());
    printf("Child PID = %d\n", pid);

    wait(NULL);
}

return 0;
}

root@LAPTOP-RJFD8EBT:~/OSSP/src# gcc main.c -o ../bin/my_shell
root@LAPTOP-RJFD8EBT:~/OSSP/src# ls -l ../bin/my_shell
-rwxr-xr-x 1 root root 16860 Aug 23 18:06 ../bin/my_shell
root@LAPTOP-RJFD8EBT:~/OSSP/src# ../bin/my_shell
my_shell> ls
This is the Parent Process.
This is the Child Process.
Parent PID = 1262
Child PID = 1263
Child PID = 1263
builtin.c executor.c main.c parser.c shell.c
my_shell> pwd
This is the Parent Process.
Parent PID = 1262
Child PID = 1264
This is the Child Process.
Child PID = 1264
/root/OSSP/src
my_shell> date
This is the Parent Process.
Parent PID = 1262
Child PID = 1265
This is the Child Process.
Child PID = 1265
Sun Aug 23 18:07:45 UTC 2026
my_shell> exit
root@LAPTOP-RJFD8EBT:~/OSSP/src#
```